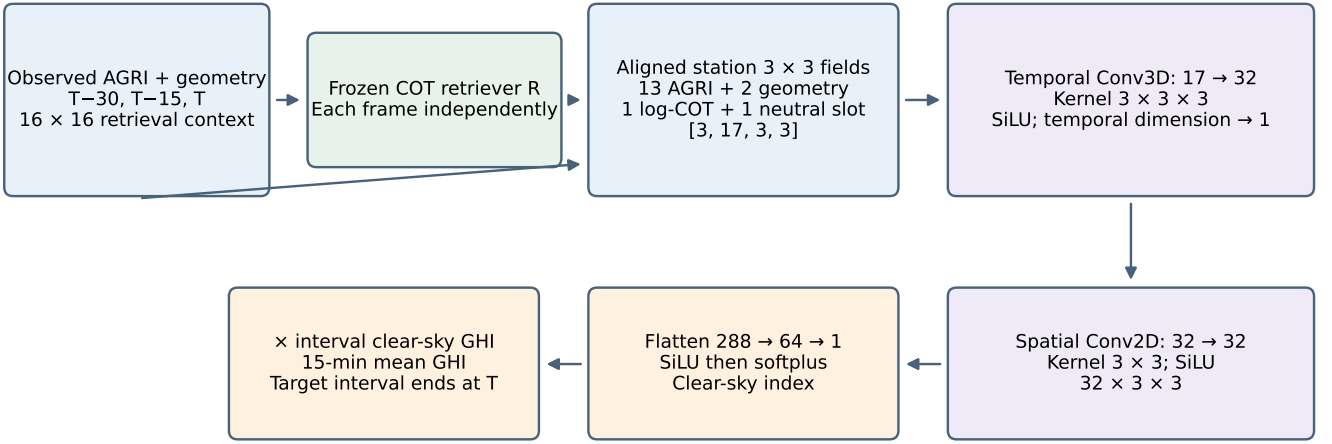

Paired comparison: same observed frames, geometry, weights at initialization and training budget.
The no-COT arm sets only the standardized COT input to zero. R remains frozen.